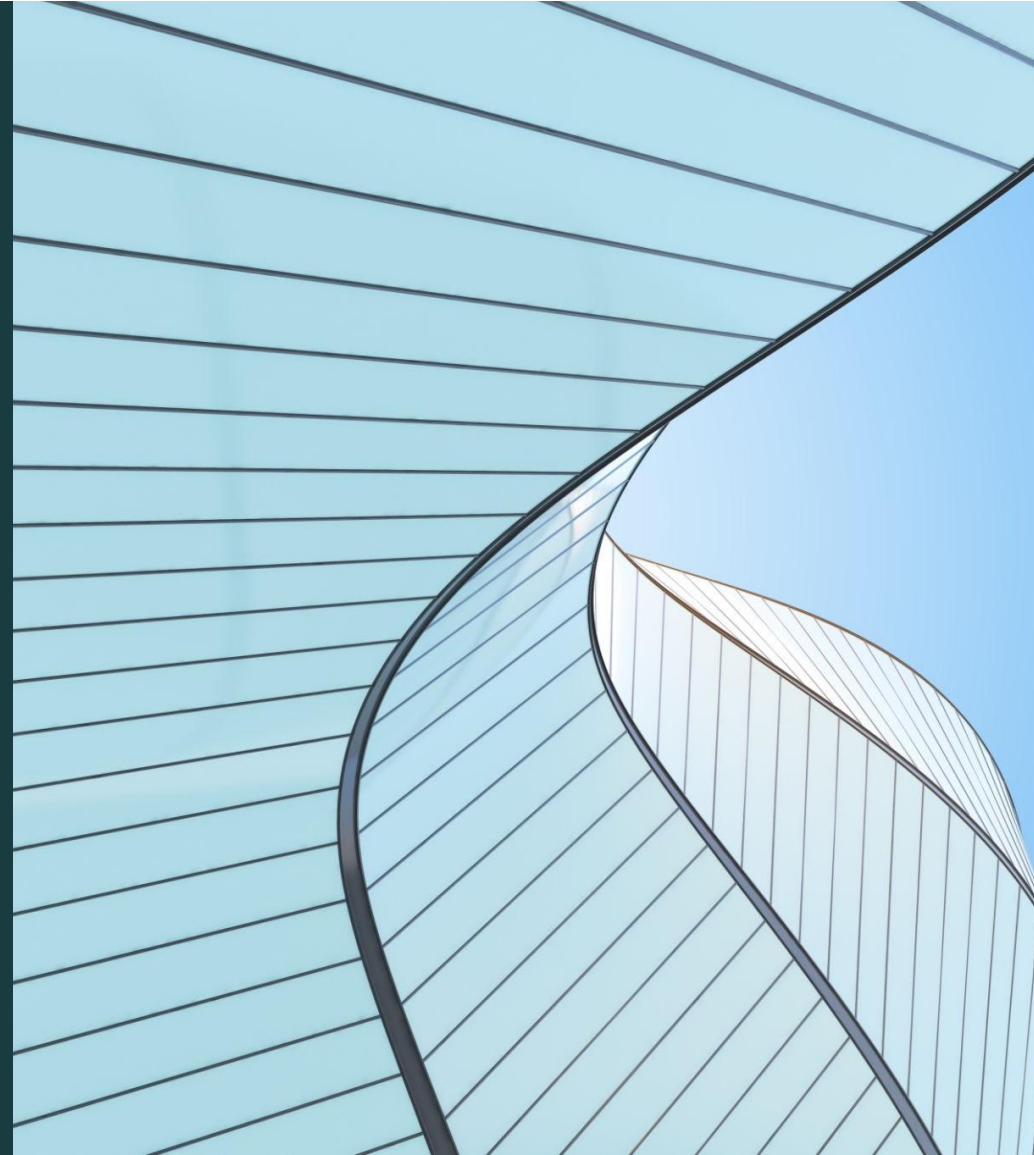


PSTAT 100 Final Project

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Topic of interest and Research question:

Which MLB games provide the best balance of affordability and fan experience for the 9/9/9 challenge?

Motivation:

We are MLB fans who want to enjoy entertaining baseball games at reasonable costs. As fans we want to find the lowest cost of these, while still watching an entertaining game in order to get the most out of our money spent.



Context:

The 9/9/9 challenge is consuming 9 beers, 9 hotdogs, in 9 innings, one of each per inning.

Goal:

Create a normalized index with mean of 100 and standard deviation of 10, to find the best games in the context of the 9/9/9 challenge as well as predict the optimality of future games. The index will take into account the price of concessions, the odds of the home team winning, and whether any 'superstars' are scheduled to play.

Methods

As a result, game enjoyability, team performance, and making an optimized guess to predict the best experience you can get from a MLB game

An entertaining game consists of **high scores, home runs, good players, and a successful team.**

We will be looking at and analyzing the win rate of teams, average scores, average hits, and other statistics in order to determine how entertaining a team is. We will then compare it to the local parks concession and ticket price costs to see if the games are worth watching per money spent.

Further Context Details

The games enjoyment will be rated with a Game Quality Score: starting pitcher quality / quality start prob, team home runs, runs, run differential, and win %



Late Quarter drawbacks:

We introduce a post-cutoff staleness penalty based on the expected duration of innings eight and later. Longer post-cutoff periods reduce the 9/9/9 optimality score, representing warming beer and cooling food.

Data Overview

- Concession prices were selected because the 9/9/9 challenge directly depends on the cost of: 9 beers 9 hot dogs
- Ticket prices were excluded from the current model because reliable historical primary-market game-level ticket prices were not available.
- MLB StatsAPI was used for game-level and player-level data because it provides consistent data across teams and seasons.

Variable Name	Measurement
concession_999_cost	Total cost of 9 beers and 9 hot dogs
game_quality_raw	Exciting/successful the game was for that team
lineup_superstar_factor_raw	Power from actual starting lineup
current_season_leader_factor_raw	Currently near league leads in HRs, hits, strikeouts, or saves
staleness_raw	Late-game waiting time after the 7th inning
999_optimality_index	Normalized 9/9/9 score

Data Data Cleaning And Preparation

Main Cleaning Decisions:

- The unit of analysis is a team-game, so each MLB game has one row per team.
- Team names were standardized across sources.
- Postponed/incomplete games were removed.
- Concession prices were merged by team-season.
- Actual starting lineups were used for the superstar factor.
- Current-season leader variables are lagged through the day before the game to avoid leakage.

Superstar Factor

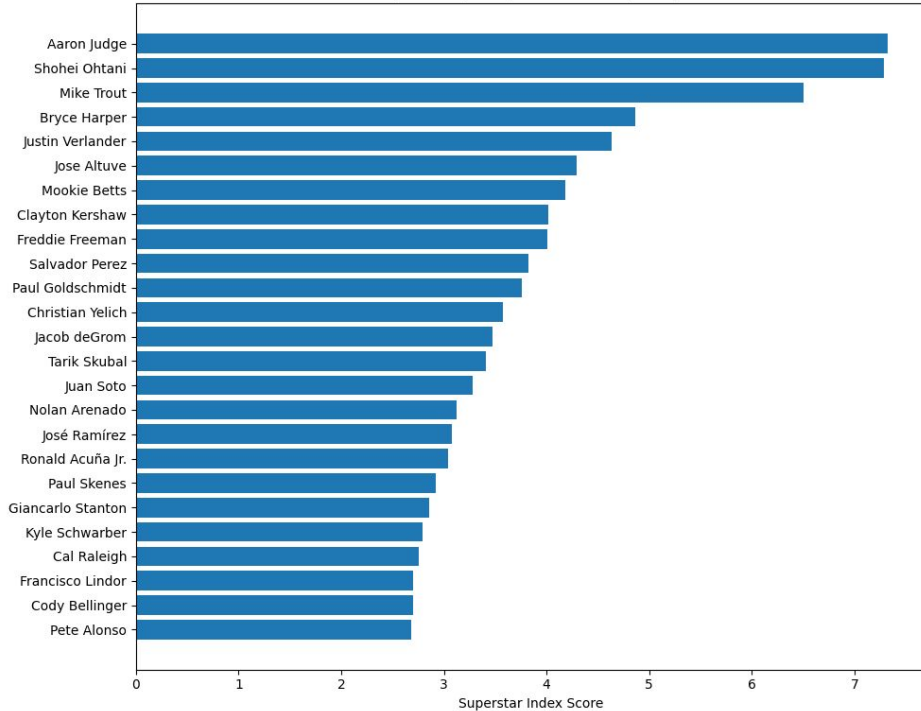
Created using:

- prior-season player performance
- career awards
- actual starting lineups

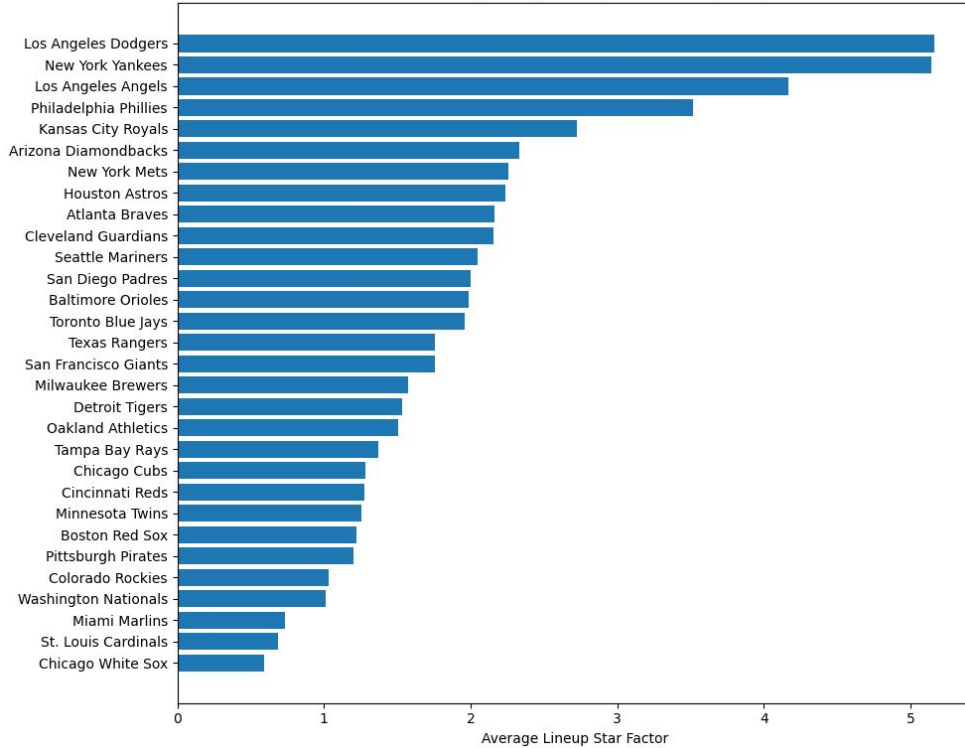
Awards were weighted by importance:

- MVP and Cy Young received the highest weights
- All-Star, Gold Glove, Silver Slugger, and similar awards received smaller weights

Top 25 Player Superstar Scores (2026)



All Teams By Average Lineup Star Factor (2026)



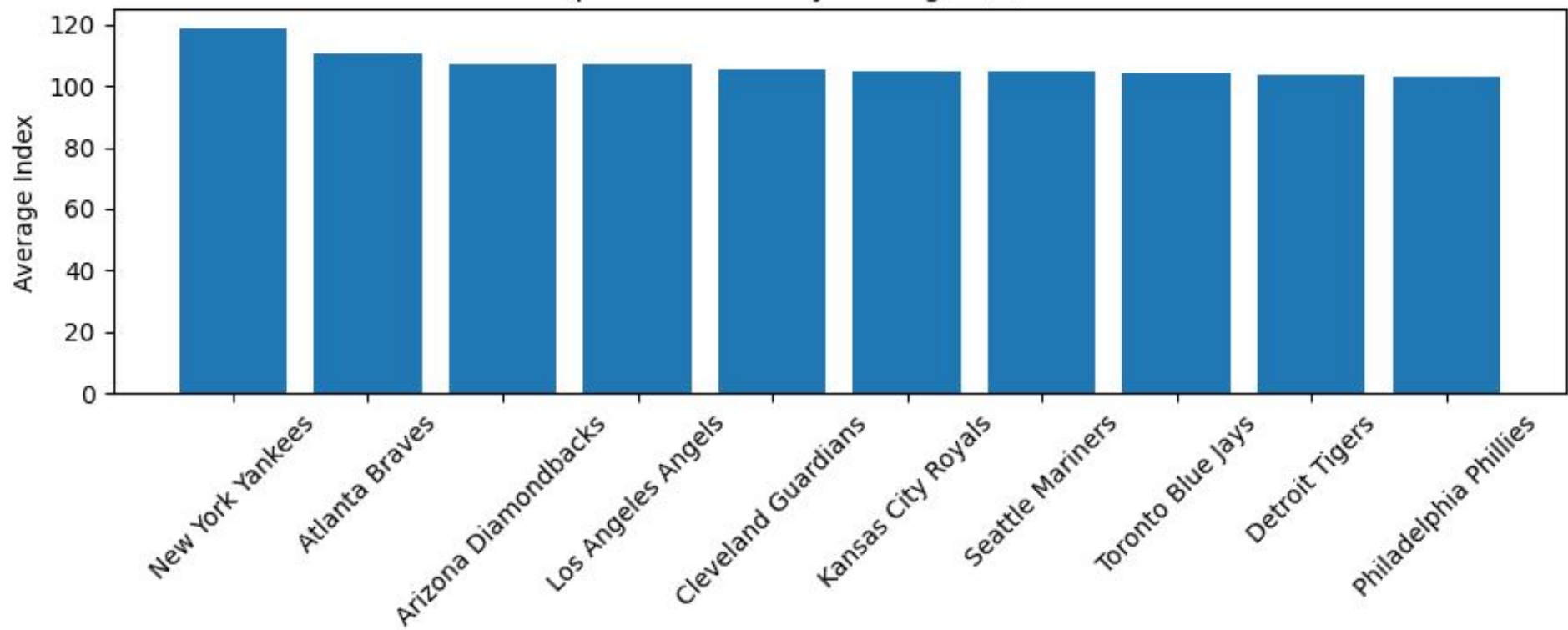
Star Factor Average

Calculated by Adding the Star Factor of each member of a team, then dividing by number of players

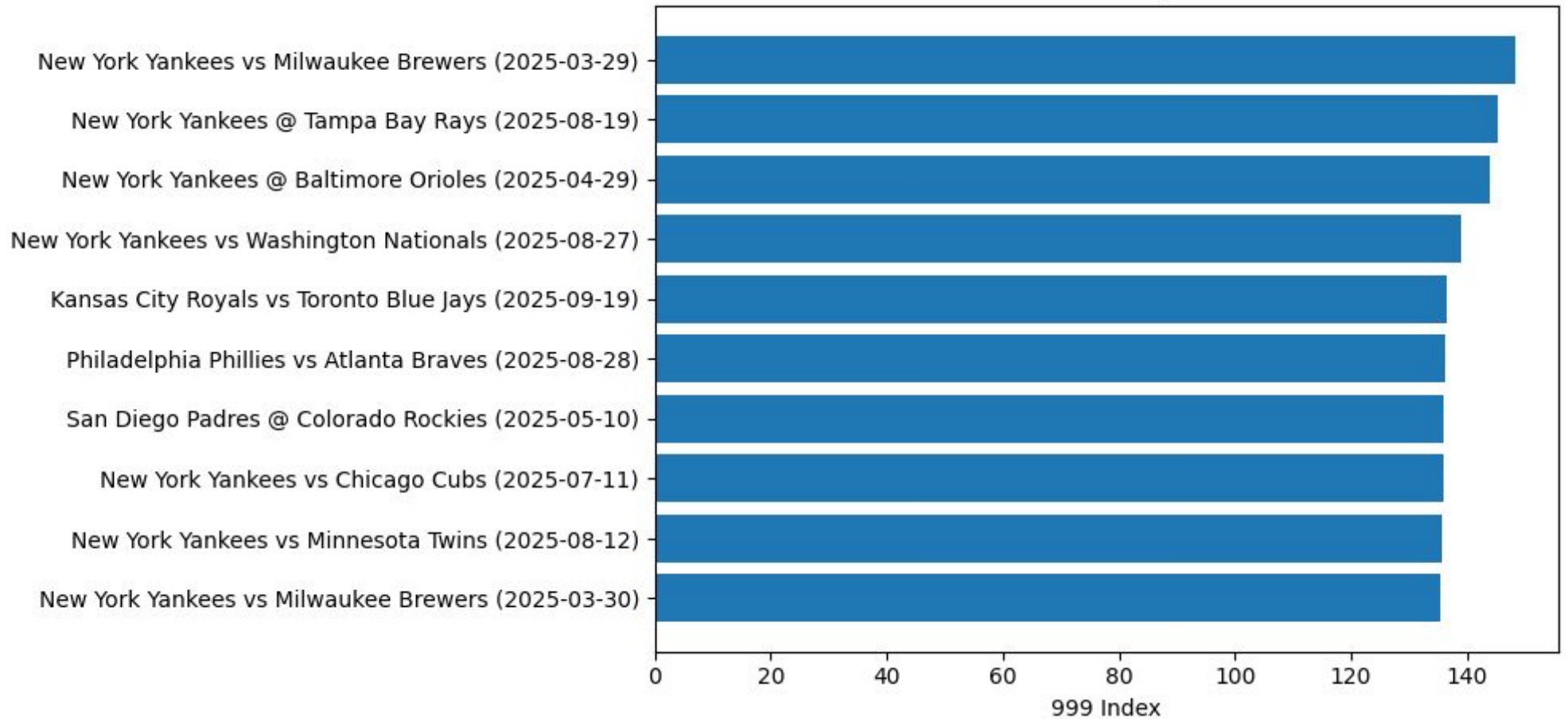
A current-season leader factor was added for players near the league lead in:

- home runs
- hits
- pitcher strikeouts
- saves

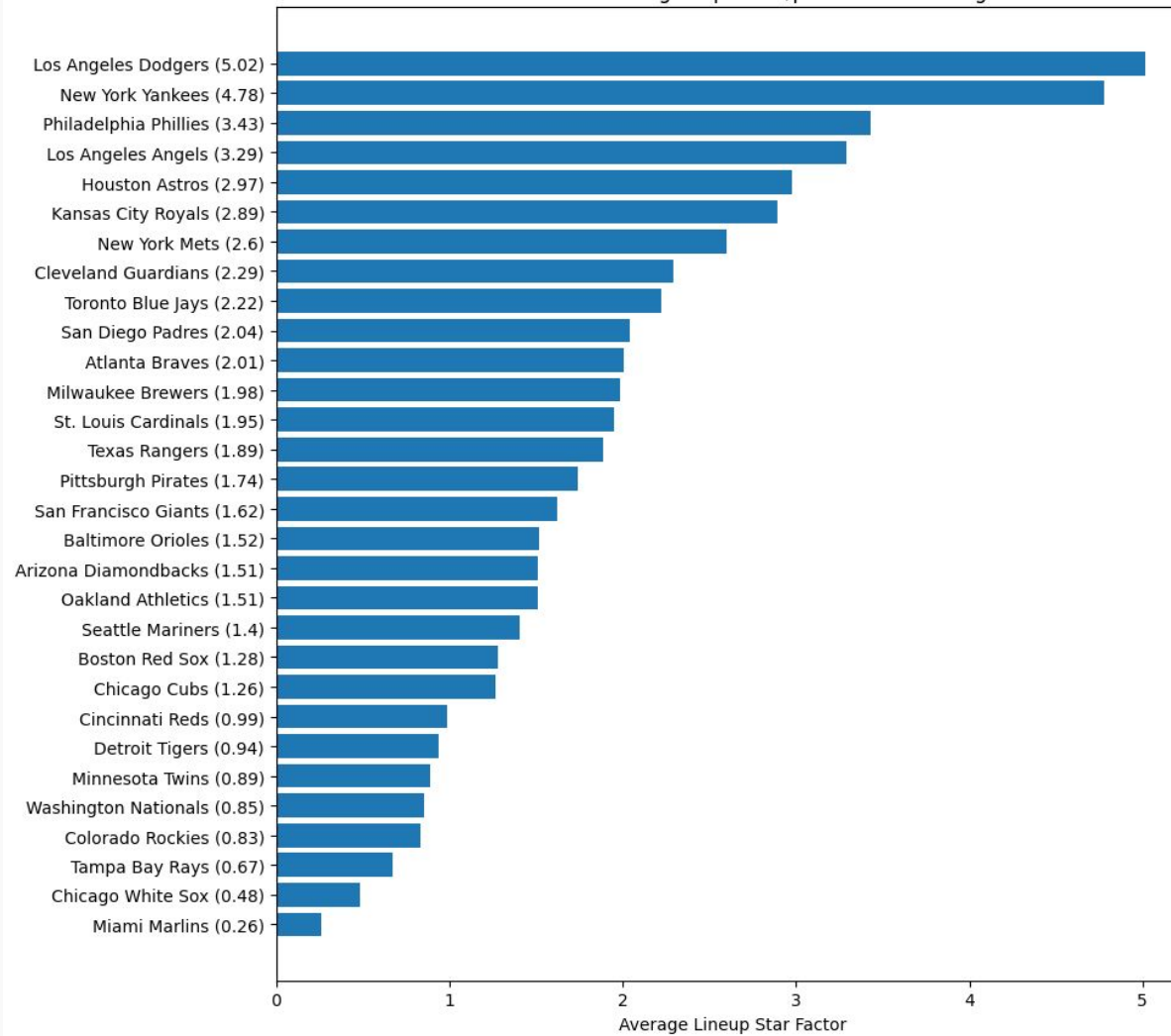
Top 2026 Teams By Average 9/9/9 Index



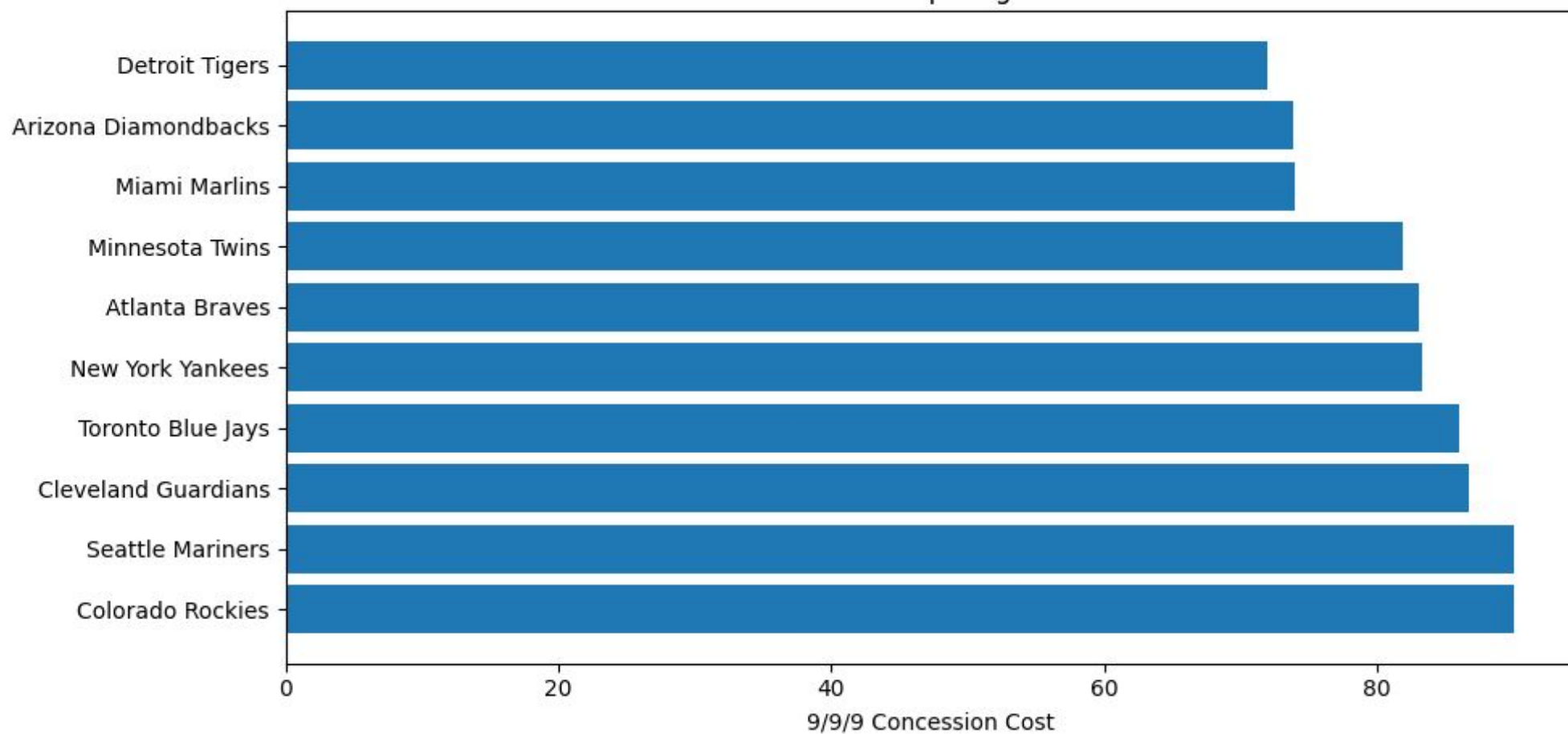
2025: Best overall game score



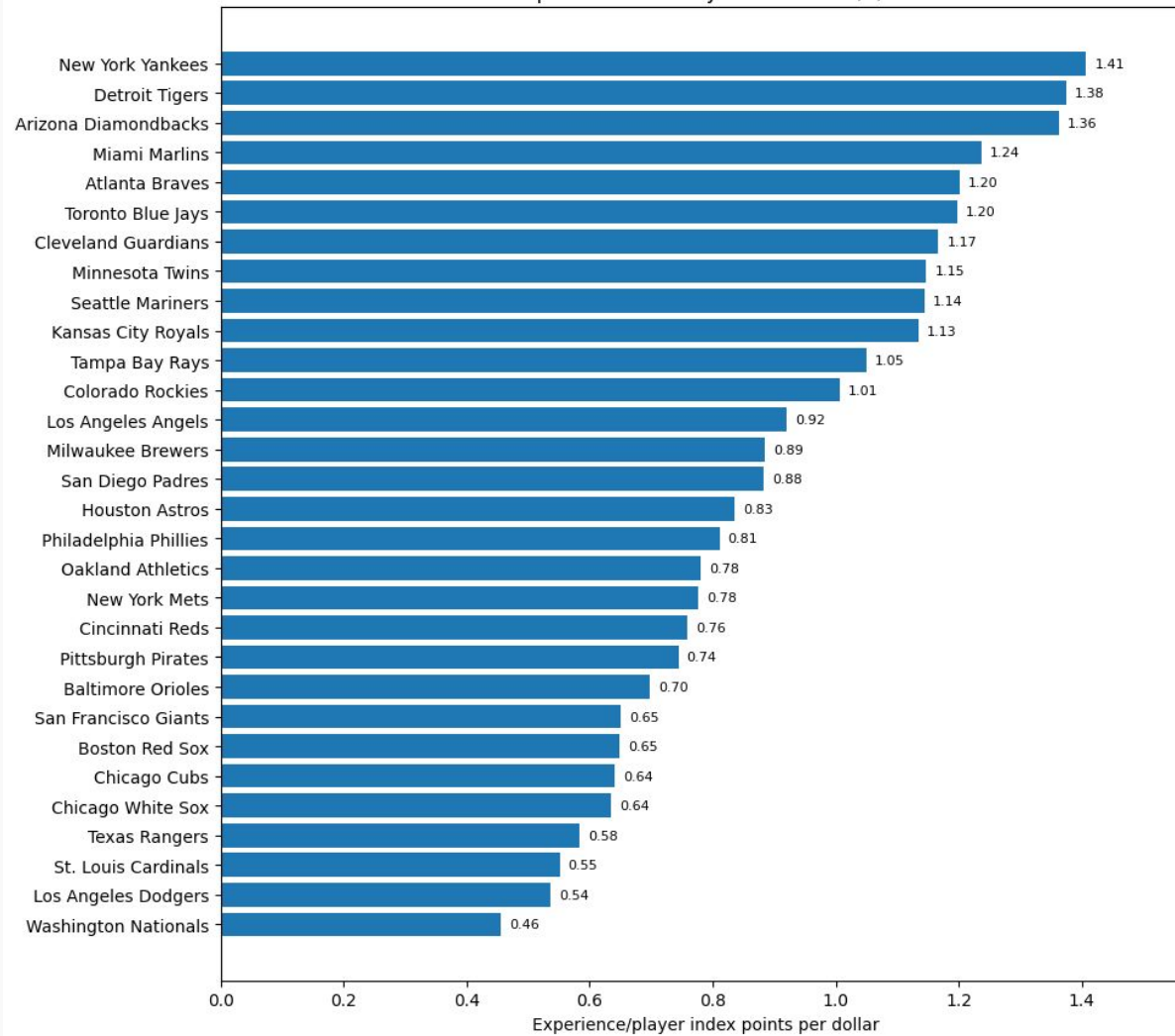
2025: Team average superstar/performance rating



2025: Cheapest games



2025: Experience And Player Value Per 9/9/9 Dollar



Analysis of Graphs

Stand Outs

The Yankees are clear favors for the 9/9/9 Index, Overall Game Score, Average Superstar Rating, earning a high entertainment rating.

The Dodgers overall experience fall behind due to high concession pricings.



Cheap Alternative

The Detroit Tigers have the cheapest games, while also having a great 9/9/9 Index. By doing so, the Tigers have a great value per dollar ratio, and are the best budget option to watch.

